Subnet mask - Important points :

1) purpose of a Subnet mask :-

* A subnet mask divids an IP address into :

→ Network portion (Identifies the network)

→ Host portion (Identifies the device within the network).

* 1s in The subnet mask = Network bits

* 0s in The subnet mask = Host bits

Subnet mask = 4 Octets (4x8 bits = 32bit)

2) CIDR Notation :

CIDR is the shorthand form of a subnet mask.

CIDR	Subnet Mask
/8	255.0.0.0
/16	255.255.0.0
/24	255.255.255.0

3) Devices can communicate Directly only if they are in the same subnet.

→ Devices compare the network portion of their IP addresses.

if the network portions match:

* They are in the same subnet.
* They can communicate directly.

if the network portion doesn't match, vice versa.

4) Examples:

with 24/

10.1.3.11 → 10.1.3.0

10.1.4.12 → 10.1.4.0

10.1.5.13 → 10.1.5.0

10.1.2.1 → 10.1.2.0

All are different subnets, so communication fails.

5) Effect of changing the subnet mask

changing the subnet mask does not change the IP address.

It changes how the IP address is interpreted.

eg IP address = 10.1.3.11

/24

Network = 10.1.3.0

Host = 11

/16

Network = 10.1

Host = 3.11

with 16, all devices belongs to the same 10.1 belongs to the same subnet

6) Default Gateway :

* It is the router used to reach other network.

* It must be in the same subnet as the host.

* If the host can't reach its default gateway, it can't access other networks or the internet.

7) Valid subnet mask rule

A subnet mask is a 32 bit binary number where

* All 1s must come first
* All 0s must come after
* It can never switch back from 0 to 1